

Xiru Fan

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Education

UC, Berkeley

Ph.D. in Material Science and Engineering

Berkeley

Sep. 2024 - Now

Shanghai Jiao Tong University

M. E. in Mechanical Engineering

Shanghai

Sep. 2021 - Mar. 2024

Shanghai Jiao Tong University

B. E. in Nuclear Engineering and Technology

Shanghai

Sep. 2017 - Jun. 2021

Publications

- [1] **X. Fan**, C. Valenzuela, W. Zhao, Z. Chen*, D. Wang*, S. J. Mentzer*. “Stochastic simulations of self-organized elastogenesis in the developing lung”, *PLOS Computational Biology*, 19(6): p. e1011219, 2023. [\[Link\]](#)
- [2] **X. Fan**, M. Zhang, K. Zhou, D. Wang*. “Multiphysics modeling and spatio-temporal optimization of grayscale digital light processing 3D printed structures with high resolution”. *Additive Manufacturing* 99, 104659, 2025. [\[Link\]](#)
- [3] M. Zhang, **X. Fan**, D. Wang*. “Voxel design of grayscale DLP 3D printed soft robots”. *Advanced Science* 2309932, 2024. [\[Link\]](#)
- [4] Q. Yu, Z. Zhu, **X. Fan**, D. Wang*. “Integrated DLP and DIW 3D Printer for Flexible Electronics”, *ICIRA 2023*. [\[Link\]](#)
- [5] D. Wang, **X. Fan**, M. Zhang. “Six-degree-of-freedom photocuring 3D printing device and 3D printing method”, *CN Patent*, CN115503232A, filed Sep 28, 2022, and filed Dec 23, 2022. (pending) [\[Link\]](#)
- [6] D. Wang, M. Zhang, **X. Fan**. “Multi-material photocuring 3D printing device and method with super air knife assisting in cleaning”, *CN Patent*, CN115625893A, filed Oct 31, 2022, and filed Jan 20, 2023. (pending) [\[Link\]](#)

Technical Skills

Programming

Matlab, Python, C++

Professional Softwares

Origin, Abaqus, Solidworks, Rhino, Mathematica, Simulink, ParaView, Lammmps

Drawing & Typesetting

Office, Illustrator, Premiere Pro, Photoshop, L^AT_EX, Markdown

Research Projects

Research on piezoelectric ceramic 3D printing

Berkeley

Ph.D research | Advisor: Rayne Zheng, *Professor of Materials Science and Engineering Department, Cal*

Jan. 2025 - Now

Aim to fabricate high-performance architected piezoelectric ceramics for sensing and actuation

- fabricate architected piezoelectric ceramics with high powder loading, low shrinkage, and precise geometry
- customize a 3D printing system enabling high-resolution piezoelectric printing
- develop a Python-based control software for 3D printer
- mitigate particle size effect on printing by oxygen control

– Electro-momentum coupling

- inversely design sacrificial scaffolds for printed green body to ensure uniform deformation and crack free densification during thermal processing
- design and fabricate acoustic metamaterial with Electro-momentum coupling effect

– Textured ceramics

- implement field assisted seed patterning and alignment during additive manufacturing
- control grain alignment to fabricated 001 textured piezoelectric ceramics with enhanced piezoelectric performance
- design and fabricate focused ultrasound transducer for medical therapy

Research on polycatenated architected materials(PAM)

Berkeley

Ph.D research | Advisor: Rayne Zheng, *Professor of Materials Science and Engineering Department, Cal*

Sep. 2024 - Now

Aim to develop responsive functional metamaterial for aerospace

- fabricate PAM with thousands of unit cell with Large area projection micro stereolithography (LaPusl) system
- measure the mechanical performance of PAM structure in liquid environment

Research on digital light processing (DLP) 3D printing

Shanghai

Master Thesis | Advisor: Dong Wang, *Associate Professor of School of Mechanical Engineering, SJTU*

Sep. 2022 - Now

Aim to develop a holistic computational tool which enables the high-resolution fabrication of soft robots with desired performance

– System set up

- build a free-form 6-DOF DLP 3D printing system using 6-axis robot arm
- build a multi-materials DLP 3D printing system

– Multi-physics modeling

- model the DLP 3D printing process considering Gaussian beam propagation, light divergence, and the photobleaching effect of the resin
- calibrate the light field parameters of the digital light engine and the photopolymerization parameters of different resin
- establish a “grayscale value - degree of conversion - mechanical properties” relationship

– Design for DLP 3D printing

- visualize the objective structure based on the planned path and projected images
- develop a spatio-temporal optimization algorithm for high-resolution grayscale DLP 3D printing
- inversely design and fabricate high-resolution structures, such as micro-fluidic devices, lattice metamaterials, and pneumatic actuators

Research on origami-inspired pneumatic soft actuator (PSA)

Shanghai

Chun-Tsung Program | Advisor: Hesheng Wang, *Professor of Department of Automation, SJTU*

Jun. 2019 - Jun. 2021

Aim to design and fabricate origami-inspired variable-stiffness PSAs with programmable performance

- design variable-stiffness skeleton inspired by origami for PSA
- simulate the transformation of the origami-inspired skeleton
- design and fabricate PSAs with programmable performance

Research on stochastic simulations of self-organized elastogenesis

Shanghai

Collaborative Research | Advisor: Dong Wang, *Associate Professor of School of Mechanical Engineering, SJTU*; Steven J. Mentzer, *Professor at Harvard Medical School*

Sep. 2021 - May 2023

Aim to simulate the process of extracellular assembly and exploring the impact of different factors on this process using cellular automata

- model the self-organization of tropoelastin in the developing lung using cellular automata based on experimental results
- analysis the impacts of different factors on the process of tropoelastin extracellular assembly

Research on reconstruction and evaluation of 3D flow field

Shanghai

Graduation Program | Advisor: Li Yang, *Associate Professor of School of Mechanical Engineering, SJTU*

Nov. 2020 - Jun. 2021

Aim to reconstruct and evaluate the 3D flow field of a ship using cross-sections generated by neural network GAN

- reconstruct flow field of a whole ship based on cross-sections after data registration and combination
- evaluate the results of 3D reconstruction using statistical and fluid mechanics indicators
- put forward a workflow of the 3D flow field reconstruction of ships based on cross-sections

Awards and Honors

Jun. 2021	Honorary Title: “The Chun-Tsung Scholar” (0.1%)	Shanghai
Jun. 2021	Honorary Title: “Outstanding Graduate of Shanghai Jiao Tong University” (1%)	Shanghai
Oct. 2019	Honorary Title: “The Merit Student of Shanghai Jiao Tong University” (1%)	Shanghai
Jun. 2019	Award: Excellent Paper Presenter of “The 5th Annual International Conference for Students”	Shanghai
Dec. 2017	Award: First Prize in “Engineering Design Showcase, 2017 Fall”	Shanghai
Oct. 2019-2022	Scholarship: “ Rong-Chang Innovation Scholarship ” (30000 RMB per year)	Shanghai
Oct. 2018-2019	Scholarship: Second Prize of the “ NPIC Scholarship ” (6000 RMB)	Shanghai
Jun. 2020	Contest: First Prize in “The Ninth Shanghai Mechanical Engineering Innovation Competition”	Shanghai
Nov. 2018	Contest: First Prize in “The ‘Zhixing’ Shanghai Student Social Practice Project Competition ”	Shanghai
May 2018	Contest: Champion of the “ 10th SJTU Mechanical Innovation Competition for Freshman ”	Shanghai

Volunteer Services

International Academic Conferences

Presentation / Attendance

- 16th International Conference on Intelligent Robotics and Applications (ICIRA 2023), Hangzhou, Jul. 2023
- 9th International Conference on Mechanical Engineering and Automation Science (ICMEAS 2023), Xi'an, Oct. 2023

International Events

Shanghai

Volunteer

- 3rd China International Import Expo, Oct. 2020
- Shanghai Marathon 2019, Nov. 2019
- 40th Odyssey of the Mind World Final, Mar. 2019

SK Sunny Student Volunteer Organization

Shanghai

Volunteer

Sep. 2017 - Jun. 2019

- Creative Innovation Class, Minhang Central Elementary School

Rong-Chang Chu-Cai Organization

Shanghai

Member

Sep. 2017 - Now

- Social practice: Healthcare Improvement in Chinese Poverty Alleviation, Eryuan Yunnan, Jul. 2018
- Social practice: Sustainable Development of Modern Companies, Shenzhen Guangdong, Aug. 2019
- Social practice: Advanced Educational Difference between China and Singapore, Singapore, Aug. 2023

References

- Prof. Rayne (Xiaoyu) Zheng
Ph. D, Associate Professor, at Material Science and Engineering Department, UC, Berkeley, Berkeley, USA
✉ rayne23@berkeley.edu
- Prof. Dong Wang
Ph. D, Associate Professor, at School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai, China
✉ wang_dong@sjtu.edu.cn
- Prof. Steven J. Mentzer
Ph. D, Professor, at Harvard Medical School, Boston, U. S.
✉ smentzer@bwh.harvard.edu
- Prof. Hesheng Wang
Ph. D, Professor, at Department of Automation, Shanghai Jiao Tong University, Shanghai, China
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